Hongyu Xu

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Education

University of Illinois Urbana-Champaign **Aug 2023 – May 2027 (Expected)**
B.S. in Mathematics & Computer Science and Brain & Cognitive Science *Urbana, IL* *GPA: **3.91 / 4.00***

- **Relevant Coursework:** Data Structures, Algorithms, System Programming, Formal Software Development Methods, Computer Architecture, Artificial Intelligence, Linear Algebra, Numerical Methods

Skills

Languages: TypeScript/JavaScript, **Python**, **C/C++**, SQL, Rust, MATLAB, Java
Full-Stack & Backend: React, Next.js, Node.js, REST APIs, PostgreSQL, Microsoft SQL Server, pgvector, background jobs
AI & ML: PyTorch, **Transformer models**, **RNNs**, LLM APIs, retrieval-augmented generation (RAG), AI agent orchestration
Systems & Testing: Tauri, Linux, **Docker**, Git, GDB, PM2, concurrency, **unit/integration/regression testing**

Experience

Tangerine Intelligence **Mar 2026 – Present**
Co-Founder & Founding Engineer *Remote*

- Independently engineered and operated an enterprise AI platform under Tangerine's **signed commercial contract** with a **publicly listed PCB manufacturer**; delivered web and Windows software for spreadsheet, ERP, and document workflows used in production across **100+ user accounts**.
- Architected full-stack services with **Next.js, React, and TypeScript**, plus **Tauri/Rust** clients backed by **PostgreSQL, pgvector**, and **Microsoft SQL Server**; implemented background jobs, **role-based access, audit logs, automated tests**, signed updates, and context-rich issue capture.
- Designed **LLM agent orchestration** with a **plan-execute-validate** workflow that converts natural-language requests and irregular enterprise files into structured, user-confirmed tool plans across ERP queries, document retrieval, and spreadsheet operations while keeping consequential writes **deterministic and auditable**.
- Built a multi-stage **retrieval-augmented generation (RAG)** pipeline for a corpus of **6,500+ files (16.6 GB)** using hierarchical context, question classification, **hybrid retrieval**, structural filtering, deduplication, **reranking**, and evidence-sufficiency checks; recovered approximately **90% of human-cited source documents**.

COmputation and NEurodynamics (CONE) Lab, UIUC **Oct 2025 – Present**
Undergraduate Researcher | Advisor: Prof. Matthew F. Singh *Urbana, IL*

- Built reusable **MATLAB/FieldTrip** tooling for EEG-MRI forward modeling across **8 sessions from 3 subjects**, including **three-layer BEM** construction, electrode registration, cortical source models, and **leadfield generation**.
- Implemented **nearest-surface projection**, **distance-minimizing alignment**, batch execution, and interactive 3D quality control; **automatically aligned 6 of 8 sessions** and corrected **2 irregular cases** for downstream modeling.
- Implemented a **self-supervised PyTorch model** within the lab's framework, using a **Transformer EEG encoder**, **autonomous RNN**, and **fixed leadfield** for latent brain-state estimation and multi-step forecasting.
- Built **MATLAB/HDF5 loaders**, **temporal train/validation splits**, and forward-model-consistent normalization; checked **gradients and checkpoint reloads** on **synthetic EEG**.

China Southern Power Grid **Jun 2025 – Aug 2025**
Enterprise Systems Intern *China*

- Supported internal office applications in an isolated intranet environment, administering **employee access permissions** and assisting with routine server maintenance.
- Implemented employee-facing query features using provided **internal APIs** and **internal database interfaces**, integrating new functionality into existing enterprise systems.

Projects

FQC Website & Interactive Engineering Demos **fqc.co.th**

- Delivered a trilingual **Next.js/TypeScript** website for FQC with **six interactive Three.js demos** and a **12-station virtual factory tour**; implemented server-side quote/contact workflows and a **streaming AI assistant** that incorporates engineering-calculator outputs as context.

Multi-threaded Monopoly Game Engine **2025**

- Built a centralized, **server-authoritative C++ engine** that **synchronized shared state** and **serialized critical rule transitions** to prevent races under **concurrent player requests**.

Awards & Honors

- USA(J)MO Qualification – AMC 12A Distinction (**Top 1%**)
- Dean's List (multiple semesters)